

Jack Moody

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SUMMARY

Adaptable and capable Cyber Warfare Officer in the US Army. Currently serving as a company commander (senior project manager) and senior analytic support officer (data scientist III) who can lead high-impact analytic teams, shape AI/ML strategy, and enable national-level defensive cyber missions. Proven record leading 170+ personnel, managing multimillion-dollar portfolios, and delivering high-impact, data-driven solutions for global missions. Skilled at translating complex technical findings into clear decisions for commanders, foreign partners, and interagency leadership. Provides a blend of operational authority, technical depth, and strategic communication to any challenge.

Trained in physics and applied mathematics, I bring stochastic and probabilistic modeling, network science, and a decision-theoretic orientation to problems where adversarial or social dynamics generate noisy, non stationary signals. I build methods that stay calibrated, interpretable, and verifiable under distribution shift and deliberate perturbation, to support decision-makers and policymakers who must act on uncertain and incomplete evidence.

EDUCATION

University of Massachusetts, Amherst <i>B.S. in Physics, Minors in Mathematics and Military Leadership, Certificate in iCons (Renewable Energy)</i> Aug. 2017 – May 2021	Amherst, MA
Johns Hopkins University <i>M.S. in Applied and Computational Mathematics</i>	Baltimore, MD May 2021 – Dec. 2023
University of Colorado Boulder <i>M.S. in Computer Science</i>	Boulder, CO Jan. 2026 – Present

PREPRINTS

Characterizing Behavioral Deviation in Endpoint Telemetry via ATT&CK Transitions <i>Moody, J., Cruickshank, I., & Ng, L.H.X.</i>	Jun. 2026
We propose a methodology that represents endpoints as a probability distribution over MITRE ATT&CK tactic transitions under two complementary encodings, scored against a leave-one-out role-peer baseline to derive three corroborating behavioral deviation signals with bootstrap uncertainty quantification. Evaluated on a 30.5-million-event Sysmon corpus spanning 27 hosts across five operational roles, producing rankings that agree on 11 of 12 candidate-deviating hosts across encodings, identifying three distinct deviation archetypes, and surfacing a single host flagged by all three signals.	
Temporal Narrative Monitoring in Dynamic Information Environments <i>Farr, D., Prochaska, S., Moody, J., Ng, L.H.X., Cruickshank, I., Starbird, K., & West, J.</i>	Mar. 2026
A systems-oriented framework that models emerging narratives as temporally evolving semantic structures, using density-based clustering and rolling temporal linkage over sentence embeddings to track narrative persistence, drift, and lifecycle without predefined labels. Evaluated on discourse surrounding a January 2026 crisis event, achieving 91% cluster-assignment coherence under stratified human validation. arXiv:2603.17617 [cs.SI]. arXiv	

THESES

Unraveling the Threat: Interpreting Terrorist Groups via Network Analysis and Decision Theory <i>M.S. Thesis, Johns Hopkins University, Applied and Computational Mathematics</i>	Dec. 2023 Baltimore, MD
Generated synthetic adversarial networks via graphons, compared them under the Gromov-Wasserstein distance, clustered the resulting metric-measure spaces using K-means, and analyzed the clusters through a decision-theoretic framework to inform courses of action; advised by Dr. Thomas Woolf. JScholarship	
Nuclear Test Film Analysis <i>B.S. Honors Thesis, University of Massachusetts Amherst, Department of Physics</i>	Mar. 2021 Amherst, MA
Developed an image-analysis pipeline (RGB channel separation, Otsu and Yen thresholding, information-entropy smoothing) to extract ejecta trajectories, velocities, and launch angles from digitized nuclear test films; advised by Dr. Rory Miskimen.	

PUBLIC WRITING

Staying Motivated and Productive During COVID-19 as an Undergraduate Student <i>APS News, Back Page, American Physical Society</i>	June 2020 College Park, MD
Op-ed for the American Physical Society's monthly membership newsletter on research productivity strategies for undergraduates during the pandemic. APS News	

CURRENT PROJECTS

STRATA-E: Structural and Temporal Role-Aware Threat Analytics for Endpoint Telemetry

Sole developer

[Github](#)

- A hierarchical, statistically calibrated behavioral anomaly detection framework for Windows Sysmon and Windows Event telemetry. The system combines time-aware sequence modeling, hierarchical Dirichlet peer-role baselines, bootstrap-calibrated Jensen–Shannon divergence, and a four-channel corroboration gate (sequence, frequency, context, drift) to reduce millions of endpoint events to a ranked, explainable host triage list with MITRE ATT&CK annotations.

PRESENTATIONS & TALKS

- **STRATA-E: STrucural and Temporal Role-Aware Threat Analytics for Endpoint Telemetry** Poster, Military Operations Research Society (MORS) AI Workshop, [2026].
- **Unraveling the Threat: Interpreting Terrorist Groups via Network Analysis and Decision Theory** Talk, Cyber Protection Brigade (CPB) Cyber Threat Summit, [2024].
- **Unraveling the Threat: Interpreting Terrorist Groups via Network Analysis and Decision Theory** Talk, 780th Military Intelligence Brigade (Cyber) AvengerCon, [2024].

PROFESSIONAL EXPERIENCE

Company Commander (Senior Project Manager)

May 2025 – Present

Headquarters and Alpha Company, 2BN, Cyber Protection Brigade

Fort Gordon, GA

- Responsible for the manning, training, and equipping of 170 Soldiers and Civilians that support Defensive Cyber Operations (DCO) globally.
- Responsible for the accountability and maintenance of 10 million dollars worth of equipment.
- Responsible for the management and allocation of a \$25,000 training budget.

Senior Analytic Support Officer [ASO] (Data Scientist III) and Trainer

May 2023 – Present

Cyber Protection Brigade

Fort Gordon, GA

- One of five qualified Senior ASOs in the US Army.
- Led projects for malicious cyber activity identification using a variety of machine learning and analytic pipelines using host and network connection logs, utilizing unsupervised machine learning techniques on the Army's Big Data Platform.
- Responsible for the content development of basic and senior ASO training / education, a tier one work role for the National Security Agency (NSA) and USCYBERCOMMAND. Specifically focusing on content for intelligence analysis, data analysis, statistics, utilization of big data platforms, jupyter notebooks, and data science fundamentals.
- Cohort lead for teaching and training of ASO candidates, personally oversaw the training and qualification of 18 ASOs over the last year.
- Supported Job Qualification Role (JQR) development between ARMYCYBER, USCYBERCOMMAND, and NSA to develop strict requirements, standards, and proper training for candidates to fill a tier one work role.

Mission Element Lead (Project Manager)

April 2024 – May 2025

Cyber Protection Brigade

Fort Gordon, GA

- Plans, prepares, executes, and assesses Hunt Forward Operations world-wide. Responsible for leading operations, and training; consolidating gains to inform Defensive Cyber Operations (DCO) hunt best practices; managing Joint, Intelligence Community, and partner country relationships that support Operations.
- Responsible for the professional development and training of eleven personnel: one Warrant Officer, four Non-commissioned Officers, five junior Soldiers, and one Army Civilian.
- Led a 20-person cross-functional incident response team to detect and disrupt nation-state cyber activity targeting critical infrastructure and operational technology, improving foreign partner-nation defensive readiness and US readiness.

AI Task Force Project Lead and Path Finder (Project Manager and AI Engineer)

July 2024 - April 2025

Cyber National Mission Force AI Task Force

Fort Gordon, GA

- Specially selected by the two star general of the Cyber National Mission Force (CNMF) to be a Project Lead for developing AI capabilities for cybersecurity applications and operations for the CNMF AI Task Force.
- Oversaw the development, testing, implementation, and documentation of multiple LLM and machine learning-based tools to increase efficiency and accuracy of cyber security threat hunting, analyst report writing, and identification of malicious cyber activity in PCAP data.
- Oversaw the infrastructure development and maintenance for multiple portable and local GPU servers for deployment for global operations.

Battalion S2 (Security and Intelligence Production Lead)

May 2023 – April 2024

Cyber Protection Brigade

Fort Gordon, GA

- S2/ Intelligence Officer for the 2nd Cyber Battalion, CPB, consisting of 400 soldiers and DA Civilians assigned to three companies and eleven Cyber Protection Teams geographically located between Fort Gordon and Fort Meade.
- Responsible for planning, supervising, and advising the Commander on intelligence analysis and collection in support of cyber protection operations worldwide.

- Responsible for ensuring compliance with security management and force protection policies and regulations.
- Supervises a section of nine Soldiers in order to execute security management and the intelligence analysis program.

Analytic Support Officer (Data Scientist II)

April 2022 – May 2023

Cyber Protection Brigade

Fort Gordon, GA

- Plans, prepares, executes, and assesses intelligence and analytic integration for Hunt Forward Operations world-wide. Responsible for directing mission enablers operations and training; consolidating gains to inform Defensive Cyber Operations (DCO) hunt best practices; managing Joint and IC relationships that support Operations. Responsible for the professional development and training of five personnel: three Non-commissioned Officers and two Army Civilians.

Experimental Nuclear Physics Research Assistant

January 2018 – May 2021

UMass Amherst, Thomas Jefferson National Lab

Amherst, MA

- Assist in and led the construction of eight Multi Wire Proportional Chambers (MWPC) for cosmic ray detection in relation to the pion polarizability experiment at the Thomas Jefferson National Lab, Newport News, VA
- Led an individual project in the design and construction of multiple miniature MWPCs utilizing 3D printing and printed circuit board design techniques for rapid prototyping

Nuclear Weapon Effects Research Assistant

May 2020 – May 2021

Defense Threat Reduction Agency & Leidos

Reston, VA

- Led an independent research project analyzing films of nuclear detonations conducted at the Nevada Test Site utilizing a variety of coding languages such as Python and FORTRAN and image analysis packages on high powered computers
- Developing communication products to educate fellow undergraduates on my research and other aspects of nuclear science research and policy for my undergraduate thesis

APS/SPS Careers Program Intern

June 2020 – August 2020

American Physical Society

College Park, MD

- Revitalized the [Physicist Profiles](#) web page to create a new engaging interface for readers to interact with based off of their career goals. This included the organization and categorization of over 60 interviews
- Assisted in the development and writing of the [Shape Up Career Series](#) designed to help physicists from all backgrounds find out what their career goals are and how to find they resources they need to succeed; complimented by this webinar I made: [APS Summer Webinar](#)

TEACHING EXPERIENCE

Graduate Teaching Assistant (Stochastic Differential Equations)

May 2023 – Present

Johns Hopkins University

Baltimore, MD

- Oversaw the grading, advising, and coaching of 20 graduate mathematics student's problem sets, exams, and independent projects over two semesters. The course teaches basic theory of stochastic differential equations, including weak and strong approximation, efficient numerical methods and error estimates, the relation between stochastic differential equations and partial differential equations, Monte Carlo simulations with applications in financial mathematics, population growth models, parameter estimation, and filtering and optimal control problems

Graduate Teaching Assistant (Theory of Statistics I and II)

April 2024 – May 2025

Johns Hopkins University

Baltimore, MD

- Oversaw the grading, advising, and coaching of 26 graduate mathematics student's problem sets, break out sessions, and exams. The topics covered include: Probabilistic models, goodness of fit tests, discrete and continuous random variables, expectation, variance and covariance, data reduction and summarization, Bayes theorem and estimators, marginals, conditionals and independence, statistical determination of models, hypothesis testing and needed tests, probability inequalities and convergence of random variables, delta methods, acceptance sampling, Poisson recursion, empirical distribution functions, negative binomials, confidence intervals, point estimates, confidence sets, method of moments, factorization theorem, order statistics, bootstrap methods, parametric inference, Bayesian inference and logistic regression. This course is a rigorous treatment of statistics that lays the foundation for advanced courses in statistics.

Graduate Teaching Assistant (Probability and Stochastic Processes II)

January 2024 – May 2024

Johns Hopkins University

Baltimore, MD

- Oversaw the grading, advising, and coaching of 48 graduate mathematics student's problem sets, break out sessions, and exams. This course teaches Gaussian random vectors and processes, renewal processes, renewal reward process, discrete-time Markov chains, classification of states, birth-death process, reversible Markov chains, branching process, continuous-time Markov chains, limiting probabilities, Kolmogorov differential equations, approximation methods for transition probabilities, random walks, and martingales

Teaching Assistant (Conceptual Physics)

January 2021 – May 2021

UMass Amherst

Amherst, MA

- Assisted in teaching and grading a 121 person conceptual physics class for non-science majors in concepts ranging from kinematics to E&M, relativity, and nuclear physics

Teaching Assistant (Electronics for Scientists)

August 2019 - December 2019

UMass Amherst

Amherst, MA

- Assisted 25 physics graduate and undergraduate students in understanding electronics systems in an open lab setting
- Taught students' experimental design techniques and problem-solving skills for electronic systems and circuits

Teaching Assistant (Intro to Physics I)

January 2018 – March 2018

UMass Amherst

Amherst, MA

- Assisted over 60 non-physics majors in understanding introductory physics concepts such as kinematics, forces, and entropy
- Taught students basic problem-solving techniques for successful completion of problem sets

ACHIEVEMENTS/ AWARDS

Order of the Sphinx Award

January 2024

Military Intelligence Corps Association

Fort Huachuca, AZ

- For exceptional contributions and unwavering dedication to advancing military intelligence in the cyber domain

Forsythe Grange Scholar Award

May 2019 and May 2020

UMass Amherst College of Natural Sciences

Amherst, MA

- A year-long \$10,000 academic fellowship that allows the recipient to advise and mentor up to four fellow underprivileged first-generation STEM students that are entering freshman

Minuteman Scholarship Award

May 2020

UMass Amherst Minuteman ROTC Battalion

Amherst, MA

- For excellence in leadership capability and commitment to leadership in academic, military, and extra curricular activities

Morton and Helen Sternheim Undergraduate Award

May 2018

UMass Amherst Physics Department

Amherst, MA

- For excellence in educational outreach and/or teacher preparation

ADDITIONAL EXPERIENCE / VOLUNTEERING

Bridge Builder Community Youth Advisory Board Member

January 2023 - Present

Bridge Builder Community

Augusta, GA

- Serve in an advisory and volunteer capacity to the Bridge Builder Communities leadership. This nonprofit provides housing, education, and vital resources to 18 - 25 year old's who are aging out of the foster care system - providing them a safe and supportive space to transition successfully into adulthood.

Johns Hopkins Engineering for Professionals Student Advisory Board

July 2021 - December 2023

Johns Hopkins University

Baltimore, MD

- Serve in an advisory capacity to the Engineering for Professionals administration by reviewing proposals and making recommendations regarding both academic and student services issues on behalf of fellow students

iCons Alumni Association

June 2021 - Present

UMass Amherst

Amherst, MA

- Coordinate events to help current iCons undergraduates at UMass develop resume and career finding skills along with participation in yearly interviews for selection of new candidates into the yearly iCons cohort.